Here are the most commonly used **Linux network commands** with their purpose and examples.

Command	Purpose	Example
ip	Displays and configures network interfaces, IP addresses, and routes	ip addr show
ifconfig *(deprecated)*	Displays or configures network interfaces	ifconfig eth0
ping	Checks connectivity between hosts	ping google.com
traceroute	Shows the route packets take to a destination	traceroute google.com
tracepath	Similar to traceroute but does not require root privileges	tracepath google.com
netstat *(deprecated)*	Displays network connections, routing tables, and statistics	netstat -an
ss	Displays socket and network connection information	ss -tuln
hostname	Displays or sets the system hostname	hostname
hostnamectl	Manages hostname and system information	hostnamectl set-hostname mypc
nslookup	Queries DNS servers	nslookup google.com
dig	Provides detailed DNS information	dig google.com
host	Performs DNS lookups	host google.com
arp	Displays and modifies the ARP cache	arp -a
ip neigh	Shows neighbor (ARP) table	ip neigh show
route *(deprecated)*	Displays or modifies the routing table	route -n
ip route	Displays or modifies routing information	ip route show
curl	Transfers data from or to a server	curl https://example.com

Command	Purpose	Example
wget	Downloads files from the web	wget https://example.com/file.zip
scp	Securely copies files between systems	scp file.txt user@host:/home/user
ssh	Secure remote login	ssh user@192.168.1.10
ftp	File Transfer Protocol client	ftp ftp.example.com
telnet	Tests TCP connectivity (legacy remote login)	telnet 192.168.1.10 80
nc (Netcat)	Reads and writes data across network connections	nc -zv 192.168.1.10 22
nmap	Network discovery and port scanning	nmap 192.168.1.1
tcpdump	Captures and analyzes network packets	sudo tcpdump -i eth0
ethtool	Displays or changes Ethernet device settings	ethtool eth0
iwconfig	Configures wireless interfaces (older tool)	iwconfig
nmcli	Command-line tool for NetworkManager	nmcli device status
systemctl	Manages networking services	sudo systemctl restart NetworkManager

Frequently Used Commands

1. Display IP Address

ip addr show

2. Check Network Connectivity

ping 8.8.8.8

3. Display Routing Table

ip route show

4. Show Active Connections

ss -tuln

5. Check DNS Resolution

nslookup www.google.com

6. Trace Packet Route

traceroute google.com

7. Secure Remote Login

ssh username@192.168.1.100

8. Scan Open Ports

nmap 192.168.1.100

9. Download a File

wget https://example.com/file.pdf

10. Capture Network Packets

sudo tcpdump -i eth0

Common Interview and Exam Commands

- ip addr – Display IP addresses.
- ip route – Show routing table.
- ping – Test network connectivity.
- traceroute – Display packet path.
- ss – Show network sockets and ports.
- nslookup / dig – DNS lookup.
- hostname – Display system name.
- netstat – Display network statistics (legacy).
- ssh – Remote login.
- scp – Secure file transfer.
- curl / wget – Retrieve data from web servers.
- nmap – Scan hosts and ports.
- tcpdump – Analyze network traffic.